

Answer key ISU-18, ISU-19, ISU-20

Source: Solution and Answer Guide, Silberstein/Obrzut, The Complete HVACR Manual 2E, Section 13: Installation and Start-Up (ISU), pages 27-28.

Exercise ISU-18

Question ISU-18.1

What pulls the refrigerant to the sensor in a halide leak detector?

Model answer: The primary air for the torch.

Question ISU-18.2

What is the normal color of the gas flame in a halide leak detector?

Model answer: Blue.

Question ISU-18.3

What is the color of the gas flame in a halide leak detector when refrigerant is present?

Model answer: Green.

Question ISU-18.4

Can a halide leak detector be used to detect natural gas? Why?

Model answer: No, an explosion would occur because the halide detector has an open flame.

Question ISU-18.5

What does the term halide refer to?

Model answer: The halogenated hydrocarbon series of refrigerants.

Question ISU-18.6

Name two gases that are commonly used to produce the flame in a halide leak detector.

Model answer: Acetylene and propane.

Question ISU-18.7

Describe how a halide leak detector discovers leaks.

Model answer: A refrigerant sample from the leak is pulled into the sample tube by the primary air for the burner. The sample of refrigerant passes by a glowing copper element and is broken down by the heat, which changes the flame color to green.

Question ISU-18.8

Does the halide leak detector work best in the light or in the dark?

Model answer: Dark, because you can see the blue flame better.

Question ISU-18.9

Why will a halide leak detector not be able to locate a leak on an R-410A air-conditioning system?

Model answer: The halide leak detector relies on the presence of chlorine to change the flame color. R-410A is an HFC blend and contains no chlorine.

Question ISU-18.10

How small of a leak can a halide leak detector typically pick up?

Model answer: As small as about 1 oz of refrigerant per year.

Exercise ISU-19

Question ISU-19.1

Can natural gas leaks be located with an electronic leak detector?

Model answer: No, ignition may occur.

Question ISU-19.2

How can the refrigerant be moved from the point of the leak to the detector sensor?

Model answer: With a small pump to provide a suction in the electronic leak detector.

Question ISU-19.3

What can be done to keep wind from affecting leak detection?

Model answer: Shield the area where you are attempting to find a leak.

Question ISU-19.4

What is the power source for an electronic leak detector?

Model answer: Batteries or 115 V AC.

Question ISU-19.5

How can soap bubbles be used to further pinpoint the leak when it is detected?

Model answer: Soap bubbles will form a bubble at the point of the leak.

Question ISU-19.6

What is the purpose of the filter under the tip of the sensor on the detector?

Model answer: To prevent airborne particles from being sucked into the sensing element.

Question ISU-19.7

Does the electronic leak detector work best in a light or dark place?

Model answer: Either.

Question ISU-19.8

How can an electronic leak detector be used to pinpoint the source of a large leak?

Model answer: Most models have a sensitivity adjustment that can be used when there is a large leak. The sensitivity can be turned down to prevent the alarm from going off before it reaches the source of the leak.

Exercise ISU-20

Question ISU-20.1

After a hermetic motor burn, what contaminant might be found in the recovered refrigerant?

Model answer: Acid.

Question ISU-20.2

What would the result be if a recovery cylinder were filled with liquid and allowed to warm up?

Model answer: The pressure would increase and it could possibly explode.

Question ISU-20.3

Why must liquid and vapor refrigerant be present for the pressure/temperature relationship to apply?

Model answer: There must be liquid present to boil if the temperature rises.

Question ISU-20.4

If an air-conditioning unit using R-22 were located totally outside and the temperature were 85°F, what would the pressure in the system be with the unit off? (Hint: It would follow the P and T relationship.)

Model answer: 155.7 psig.